

Inventory_based_disaggregation.R

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disaggregation	<i>Disaggregate emissions to the pixel scale using ACES or Vulcan CO2 inventories</i>
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Description

This is a function utilized by ‘stationary_combustion’ and ‘NG_distribution’. It distributes emissions from a larger scale (state or county) to the pixel scale using gridded CO2 emissions.

Usage

```
disaggregation(  
  input_inventory,  
  totals,  
  agg_level,  
  NEI_input,  
  cover_all,  
  out_envir  
)
```

Arguments

input_inventory	SpatRaster. Either a Vulcan or ACES sectoral inventory. ACES is available at https://doi.org/10.3334/ORNLDAAAC/1943 and Vulcan is available at https://doi.org/10.3334/ORNLDAAAC/1741 . Annual mean files should be used for Vulcan. The hourly ACES file should be averaged across hours to create an annual mean inventory. There is a function to do this as well, though it is quite time consuming given the amount of data that needs to be downloaded.
totals	Character vector. Various subsectors to run through the disaggregation process.
agg_level	Character. The scale of the data before disaggregation, solely used for naming.

NEI_input	SpatVector. Includes attributes for each spatial feature that detail the emissions at the current <code>agg_level</code> separated by totals.
cover_all	List of matrices. Each entry in the list is a separate state/county and the matrices are the output of the function <code>terra::extract(weights=T,exact=T,cells=T)</code> . These provide the fraction of each pixel contained within the state/county. This allows for better accounting for pixels that are on the borders of states/counties.
out_envir	Environment. Where to assign output data. Should be the environment of 'stationary_combustion' or 'NG_distribution' as the output will be used and exported there.

Details

This function is intended to be used by 'stationary_combustion' and 'NG_distribution'. For each pixel it takes the ratio of CO2 from that pixel to the total CO2 for the state/county it is in as input. This is combined with the state/county total CH4 emissions (also input) to disaggregate the emissions to the pixel-scale. If there are no emissions for the state/county then the entire area is assigned an equal fraction of emissions. This is done using sectoral (residential, commercial, industrial, electric) CO2 inventories.

ACES is available at <https://doi.org/10.3334/ORNLDAC/1943> and Vulcan is available at <https://doi.org/10.3334/ORNLDAC/1741>.

See references [Vulcan](#) and [ACES](#)

Value

Nothing is returned. The disaggregated methane data is assigned to the specified `out_envir`. This will be a `SpatRaster` list with one for each of the totals.

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Examples

```
aces_res <- rast(paste0(ACES_directory,"/Sectoral/",ACES_year,'_Annual_ACES_Residential.nc'))
res_totals <- c('mains_ER_total_res',
               'serv_ER_total_res',
               'MnR_ER_total_res',
               'meter_ER_total_res',
               'upset_ER_total_res',
               'post_meter_ER_total_res')
cover_all <- list(extract(aces_res,all_merge_LCC_domain,weights=T,exact=T,cells=T))
disaggregation(aces_res,
               res_totals,
               agg_level="state",
               NEI_input=all_merge_LCC_state,
               cover_all,
               out_envir=environment())
```

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